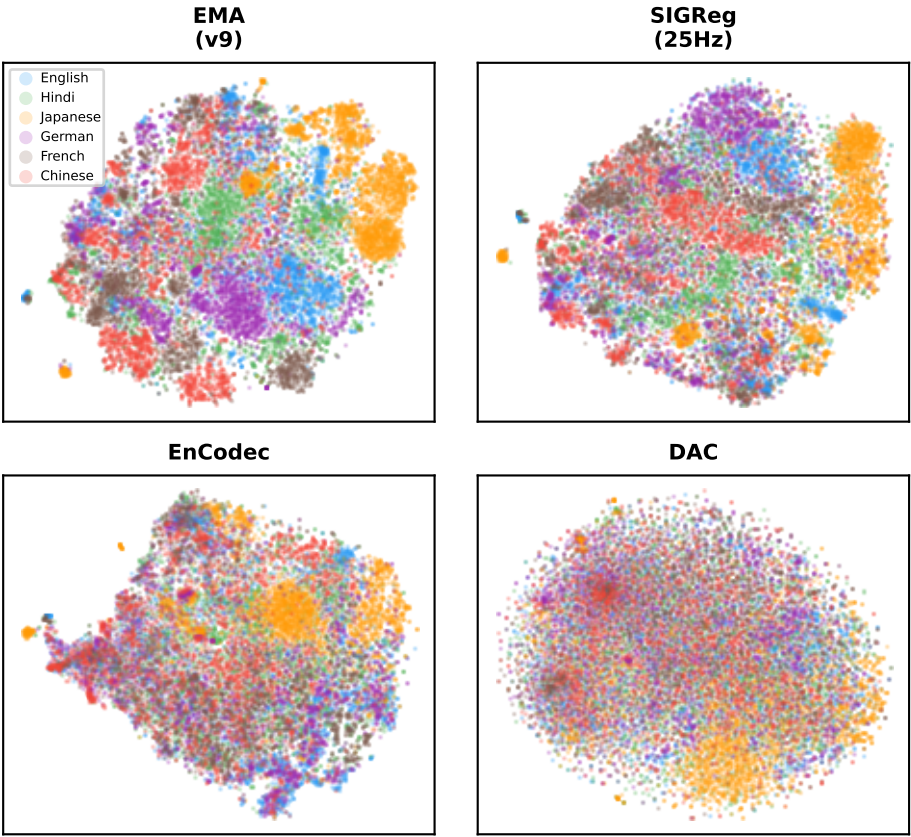
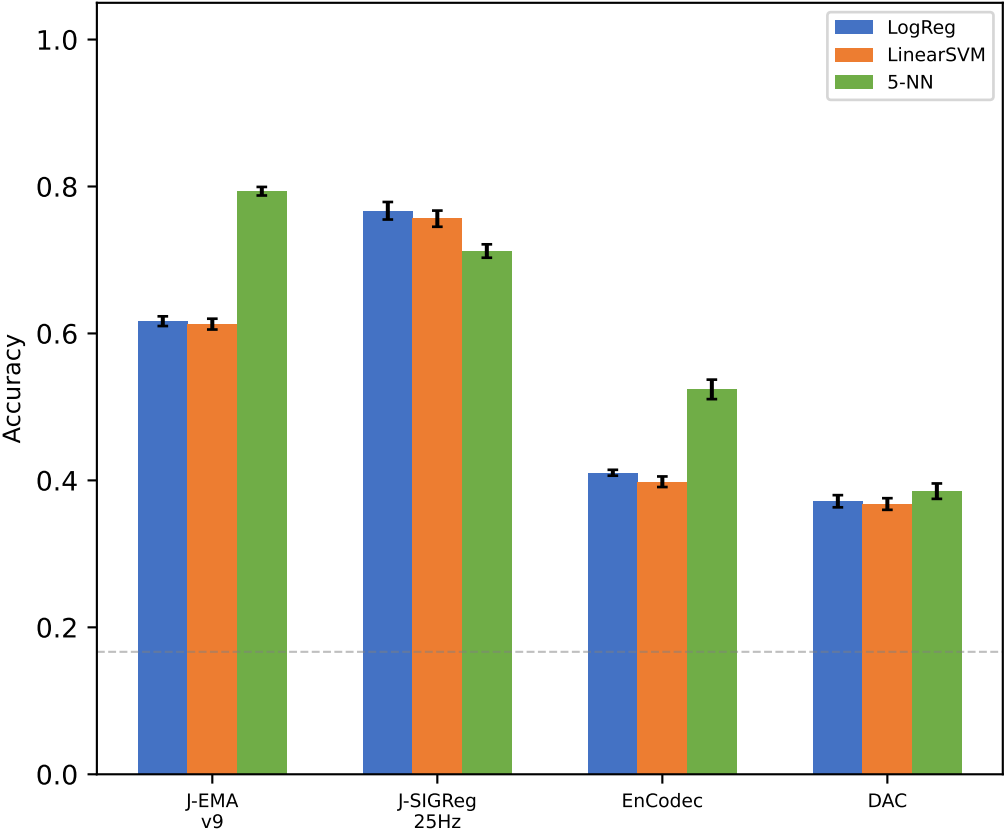


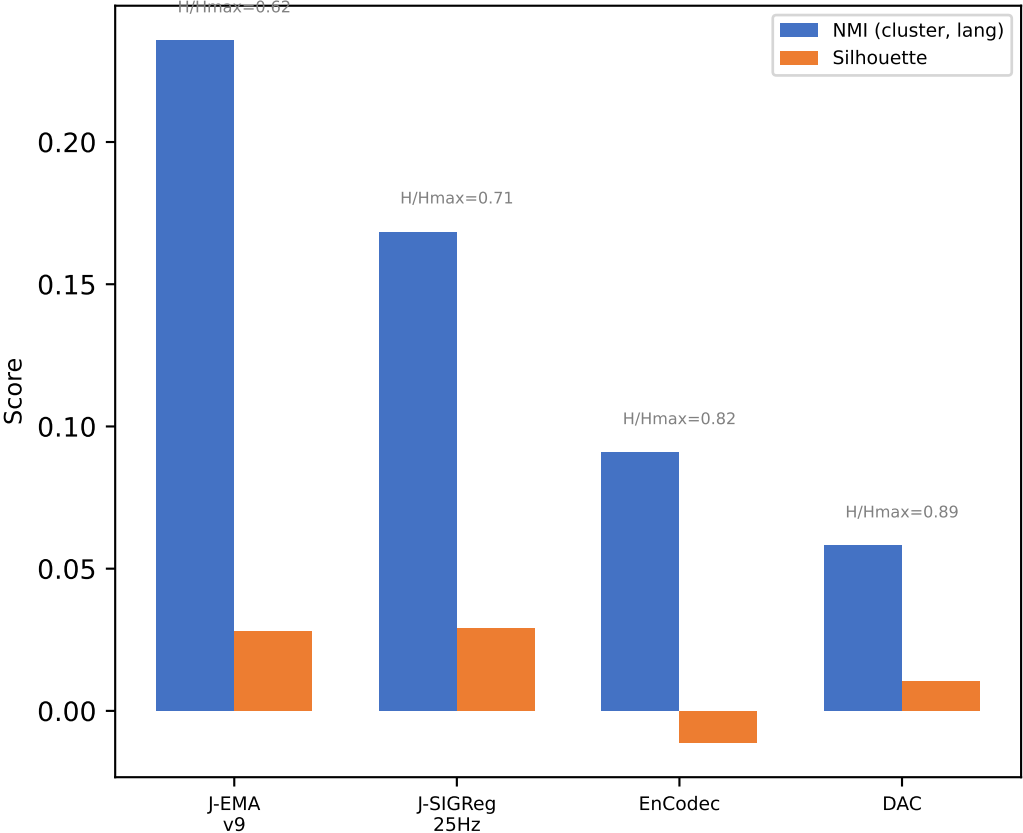
(A) t-SNE: Speech-Only Frame Embeddings



(B) Language Separability (lower = more shared)



(C) Cluster Quality (k=64)
NMI: lang correlation | Sil: cluster tightness



(D) Coarse Acoustic Categories (speech-only, k=64, top 12)

#	N	RMS	ZCR	SC(Hz)	F0	V%	Category
14	1370	0.019	0.075	1126	176	5%low-energy transiti	
8	1272	0.019	0.088	1899	211	4%	frication
22	1216	0.013	0.054	994	173	4%	sonorant/vowel
60	1194	0.022	0.075	1478	183	5%low-energy transiti	
51	1163	0.036	0.071	1165	218	1%	unvoiced/plosive
18	1146	0.023	0.065	1419	190	3%	mixed
12	1063	0.022	0.054	1117	179	15%	sonorant/vowel
30	1057	0.049	0.058	971	135	5%	sonorant/vowel
2	1022	0.029	0.073	1500	191	8%	unvoiced/plosive
5	1014	0.033	0.072	1186	229	0%	unvoiced/plosive
40	980	0.020	0.066	1087	190	8%	mixed
28	963	0.021	0.062	1052	214	10%	mixed